CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B.Tech. Examination (CE)

November 2013

CE209: Design & Analysis of Algorithm

Date: 13.11.2013, Wednesday | **Time: 01:30 pm to 04:30 pm** | **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION-I

Q-1 Answer the following questions:

(a) Give the diagram representation of Notion of algorithm. **2**

(b) Is Dynamic Programming Top Down Approach or Bottom Up mechanism? Why? **2**

(c) Explain: Big-oh notation, Big-omega notation, Theta notation. **3**

Q-2 Do as directed:

(a) For string matching working module q=11, how many spurious hits does the Rabin-karp matcher encounter in the text T=3141592653589793, when looking for the pattern P=26? **5**

(b) "Dynamic programming can be overkill while greedy algorithms tend to be easier to code". Justify the sentence. **3**

(c) Find an optimal parenthesization of a matrix-chain product whose sequence of dimension is {4, 10, 3, 12, 20, 7}. **6**

OR

Q-2 Do as directed:

(a) Show the comparisons the naïve string matcher makes for the pattern P=10001 in the text T= 00000100010010. **5**

(b) *"Explain why the statement "The running time of algorithm ABC is at least O (n^2)", is meaningless".* **3**

(c) You are given two strings s1 and s2. Write down an algorithm to computer length of the Longest Common Subsequences of given two strings. **6**

Q-3 Attempt Any Two. **14**

(a) Write Kruskal's algorithm for minimum spanning tree with an example and also explain the running time of algorithm.

(b) Explain the 4-Queen's problem & discuss the possible solutions.

(c) Using Branch & Bound Technique solve the Assignment Problem with the following cost matrix.

	1	2	3	4
A	11	12	18	40
B	14	15	13	22
C	11	17	19	23
D	17	14	20	28

SECTION-II

Q-4 Answer the following questions:

(a) Which of the following sorting algorithm does not have a worst case running time of O (n^2)? 1

a. Insertion sort b. Merge sort c. Quick sort d. bubble sort

(b) Is $2^{n+1} = O(2^n)$? Is $2^{2n} = O(2^n)$? 2

(c) Differentiate: Decision Problems and Optimization Problems. 2

(d) What is algorithmics? Give the general plan for divide-and-conquer algorithms. 2

Q-5 Do as directed:

(a) Give an algorithm to compute the Factorial of a positive integer n and analyze its efficiency. 3

(b) How to choose an algorithm for solving a single problem, When you have multiple algorithms for solving same problem are available? Which function takes less time to solve the given problem from below? Give proof for your answer. 5

$$f(n)=f(n-1)+n$$
$$f(n)=2\,f(n-1)+1$$

(c) Write down GREEDY_ACTIVITY_SELECTOR algorithm that gives an optimal solution to the activity selection problem. 6

Q-5 Do as directed:

(a) Find the worst case time complexity of following program: 3

```
bool duplicate=false;
for(int i=0;i<n;++i) {
  for(int j=0;j<n;++j) {
     if(i!=j && A[i]= =A[j]){
      duplicate=true;
      break;
     }
  }
  If(duplicate) {
    break;
  }
}
```

(b) Illustrate the operation of INSERTION SORT on the array A=<2, 13, 5, 18, 14>. Also analyze insertion sort algorithm. 5

(c) Write down an algorithm for Depth First Search of a graph? Give an example for it. 6

Q-6 Attempt Any Two. 14

(a) What do you mean by D & C strategy? Analyze the merge sort algorithm. Argue on its best case, average case and worst case time complexity.

(b) List out the methods used to solve recurrence equation. Also Solve the following recurrence:

$$F_n = \begin{cases} n & n=0,1 \\ F_{n-1}+F_{n-2} & \text{otherwise} \end{cases}$$

(c) Why do we consider Asymptotic Notations Important? Give asymptotic upper bound and lower bounds for T (n) in each of the following recurrences. Assume that T (n) is constant for n<=2. Make your bounds as tight as possible, and justify your answer.

1. $T(n)=16T(n/4)+n^2$
2. $T(n)=2T(n/2)+n^3$
3. $T(n)=2T(n/4)+\sqrt{n}$

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